

Cross Recurrence Quantification Analysis of Mild Cognitive Impairment EEG under Working Memory Condition

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Abstract— This work investigates the effectiveness of the nonlinear method of cross recurrence quantification analysis (CRQA) for classifying EEG of subjects with mild cognitive impairment (MCI) based on their synchronization levels. The synchronization levels are quantified using the CRQA recurrence rate (RR) of EEG under memory activation state of picture recollection (PICREC) for both inter and intra lobar levels. In inter lobar level, MCI EEG shows higher RR values of CRQA with statistical significance on all regions when compared to the EEG of normal control group (NC). In intra lobar analysis also, RR values of MCI are significantly higher but only in the temporal and parietal regions. Receiver operating characteristics (ROC) is used for classification of EEG of these two groups. A fair classification is obtained between the EEG of MCI and NC by using ROC analysis under this memory activation task state.

Keywords — Cross Recurrence Quantification Analysis; Receiver Operating Characteristics; Mild Cognitive Impairment; EEG.

I. INTRODUCTION

Alzheimer's disease (AD) is a disordered cognitive condition with loss of intellectual ability or thinking skill. The early symptoms of most of these disorders are memory impairment which may lead to cognitive dysfunction. Mild cognitive impairment (MCI), the early stage of AD, characterized by greater memory deficits than observed in normal elderly, but without fulfilling the clinically probable AD criteria [1]. More memory problems are present in subjects with MCI than in normal people of the same age. MCI is a condition where only memory activities are mainly affected, leaving other cognitive domains relatively less affected [2]. Currently, in clinical practice, AD diagnosis is done using brain imaging and other technical methods, which are quite expensive. As AD and MCI are cortical disorders, characteristic changes are expected to be produced in the electrocortical activities and thereby reflected in the EEG.

Features of AD/MCI EEG have been identified by both linear as well as nonlinear methods. Linear spectral features of these EEG are a decrease of the mean frequency and coherence of EEG [3, 4]. Considering the fact that EEG is a highly nonlinear signal, several attempts have been made to study its nonlinear features [5]. Such studies have identified that AD

EEG shows lesser complexity than that of normal. However, it is also observed that most of such conventional measures show sensitivity to the nonstationarity of the signal and will work effectively only on noise free long data sets. These properties are nonacceptable for any measure which is proposed for the analysis of real practical signals. Other studies using synchronisation measures have identified changes in synchronisation between different cortical regions which is the indication of damaged coordination of neuronal activities [6]. It is observed that irregularities in the synchronisation pattern is better revealed in the memory activation task states [7, 8].

Recurrence and cross recurrence methods are highly efficient for the analysis of nonlinear systems. Several measures of recurrence quantification analysis (RQA) are effectively being used in the nonlinear analysis of real world time series. These measures are extremely robust to the nonstationarity of the signal and works effectively even for short signals. Cross recurrence methods investigates the similarity of two processes as inferred from their corresponding time series. Cross recurrence quantification analysis (CRQA) provides measures for similarity of trajectories of two processes in a common phase space [9]. Several research works were done in various fields like hypnotic assessment and physiology using RQA and CRQA quantification analysis [10,11].

Here we focus on the application of CRQA method for the MCI EEG classification from the information provided by the synchronization analysis. EEG of MCI subjects and normal controls (NC) are analysed under the memory activated state of picture recollection (PICREC). In the present work, the synchronisation levels of MCI EEG are quantified using recurrence rates (RR), which is the simplest measure of CRQA. Classification between MCI and normal EEG is done using receiver operating characteristics (ROC).

II. METHODS

A. Subjects

The present study involved eighteen MCI subjects (seven females) having average age 67.1 ± 7.2 years and eighteen age matched elderly normal control (NC) subjects (five females) with average age 65.1 ± 5.6 years. All the participants are right handed people speaking the native language of

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Kerala, Malayalam. Mini mental state examination (MMSE) score is to assess the cognitive performance of the subjects in both groups of MCI is 26.2 ± 1.4 whereas it is 30 for NC. Appropriate parametric/non-parametric tests are used to assess the age, MMSE and gender composition and it is observed that the two groups do not differ significantly ($p > 0.05$), $\chi^2(16) = 15.000$ in age and gender ratio whereas the MMSE scores differ significantly ($p < 0.05$). The control subjects voluntarily participated in the study by giving informed written consent, and the same were given by care-givers of MCI subjects. The ethical committee of Welcare Hospital sanctioned the study.

Clinical assessment, neurophysiological, radiological and biochemical testing were used to perform the diagnosis of MCI. Only the memory domains were affected for this group leaving other cognitive functions unaffected. Inclusion criteria were (i) Subjective loss of memory as inferred from medical history (ii) No dementia according to the DSM IV criteria (iii) Normal performance in general cognitive activities excluding memory loss (iv) Independent daily activities (v) No psychiatric history (vi) MMSE score ≥ 24 . The exclusion criteria included (i) Cases of head trauma (ii) Seizure/epilepsy (iii) Anxiety/depression (iv) Dementia (v) Delirium (vi) Substance abuse (vii) Focal lesions in gray or white matter (viii) Psycho pharmacological treatment.

B. Experimental protocol

For the experimental purpose, the EEG of subjects from both groups are recorded under the cognitive task of picture recollection (PICREC). The subjects were made to sit in a semi reclined position in an armchair. The task was given after ensuring that they are in a relaxed position with eyes closed and with minimum movements. The subjects were shown 10 pictures on the computer screen and were asked to watch and memorize them. After 10s, they were asked to recollect as many pictures as they could remember.

C. EEG acquisition and preprocessing

Thirty two channel EEG were acquired according to the 10-20 system with the electrodes referenced to linked earlobes using Digital Wingraph EEG system (Neurocare, Model Number 13000) from the electrode sites, A1, A2, Cz, Fp1, Fp2, F3, F4, F7, F8, T3, T4, T5, T6, C3, C4, P3, P4, O1 & O2. Sampling rate of signal acquisition was 128Hz and the signals were filtered at bandpass of (0.4-60) Hz and a notch of 50 Hz. 10s epochs free from artifacts were selected from all electrodes, except Fp1 and Fp2, which are generally more affected by ocular artifacts for further off-line analysis.

D. CRQA Techniques

RR of CRQA measures how dense the state points of the two systems occupy the state space. It represents the probability of observing the two systems in the same state at the same time or at a prescribed delay [12]. It is calculated as follows.

$$RR(t) = \frac{1}{N-t} \sum_{l=1}^{N-t} lP_t(l) \quad (1)$$

where 'N' is the number of points in a time series and $P_t(l)$ is the probability of similar states to be observed in both

systems with a given delay 't'. [12]

E. EEG Analysis

Analysis using CRQA is conducted on the 10s epochs from the 14 electrode locations. Inter and intra lobar synchronization analysis of every pair of EEG signals from these 14 channels is carried out and the average value of CRQA RR is calculated. Inter lobar RR value of CRQA between any two regions is calculated by taking the average of all channel pairs between those regions. For e.g. the RR of CRQA for the temporal-parietal pair is calculated by averaging the RR values between every pair of electrodes from the two regions viz. T3, T4, T5, T6 (temporal) and C3, C4, P3, P4 (parietal). Similar analysis is carried out to obtain the synchronisation levels between all other region pairs. For intra lobar analysis, the RR value between every channel pair within a cortical region viz. frontal, temporal, parietal and occipital are averaged to obtain the corresponding synchronization levels.

F. Statistical Analysis

Multivariate ANOVA (MANOVA) is conducted using IBM SPSS for windows version 20, for a comparison of average inter and intra lobar CRQA RR between the two groups for each of the four regions. ROC analysis is used to determine the efficiency of CRQA RR in classifying the MCI and NC EEG under the cognitive state of PICREC.

ROC curve for different threshold are plotted with the true positive rate (TPR) against the false positive rate (FPR). The accuracy of the test is assessed from the value of area under this curve [13]. This quantifies the efficiency of the diagnostic test to differentiate between the two groups. A value of 1 represent a perfect test and a value of 0.5 shows that test is worthless. Inter and intra lobar synchronization levels are determined by calculating the RR values of CRQA for both groups in the PICREC state.

III. RESULTS AND DISCUSSION

The mean CRQA RR values of inter lobar analysis for both MCI and NC under the cognitive state of PICREC is shown in Fig.1a. '*' shows that statistical analysis indicates a significant difference between the two groups. Statistically significant higher values observed for all pairs of regions shows increased synchronization under PICREC cognitive state in MCI. Similarly Fig. 1b shows the mean RR values of CRQA for the intra lobar analysis under the same cognitive state of PICREC. Here also, a higher synchronization is observed for MCI in all the four regions. However, significant difference is obtained only in the cortical regions of temporal and parietal.

Higher CRQA RR values indicate the disturbed coordination between the cortical regions which can be inferred as MCI brain incorporating compensatory mechanism for completing the task. To investigate how well the CRQA analysis can classify the MCI EEG, ROC curves are drawn for inter and intra lobar analysis and these are shown in Fig. 2 and 3 respectively. Area under curve (AUC) gives the classification efficiency between the groups. The AUC values are arranged in Table I and Table II for inter and intra lobar analysis respectively.

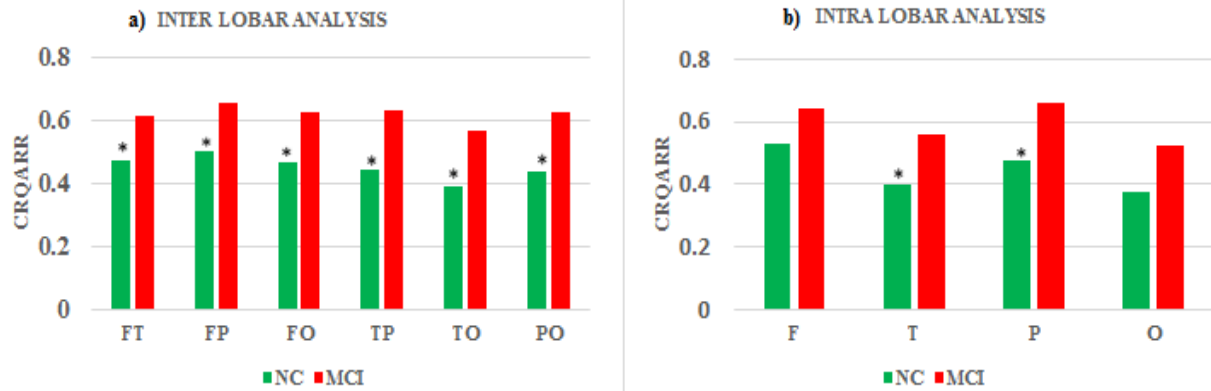


Figure 1. Average CRQA RR of a) inter b) intra lobar analysis of MCI and NC under the cognitive state of PICREC

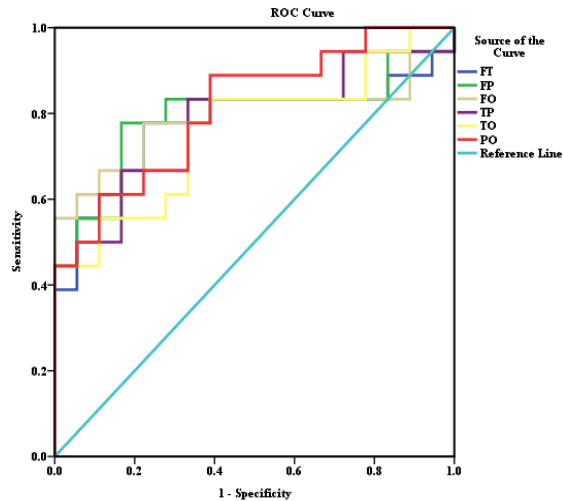


Figure 2. ROC curves of six region pairs from inter lobar analysis

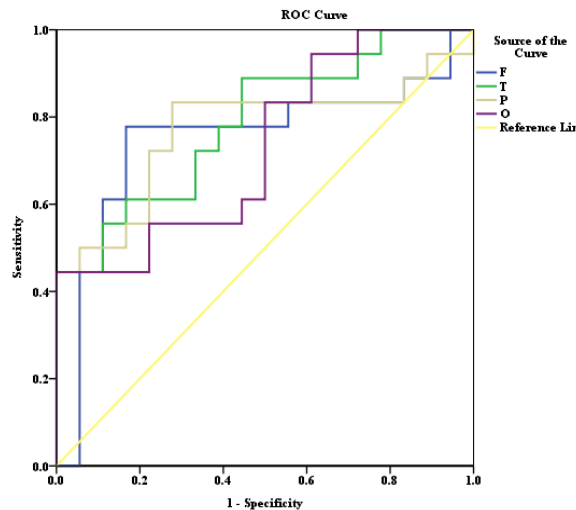


Figure 3. ROC curves of four regions from intra lobar analysis

From Table I, it is seen that highest AUC value is for parietal-occipital region pair (0.812), which is a good classification. For the remaining five region pairs the values for AUC are in between 0.759 and 0.793. Hence, a fair classification is obtained for these pairs using ROC analysis.

Similarly, it can be seen that the values for AUC lies in between 0.731 and 0.787 for all regions in the intra lobar analysis from Table II. Therefore, a fair classification of MCI EEG is obtained in all the four regions in the intra lobar analysis also.

TABLE I. AREA UNDER CURVE FOR INTER LOBAR ANALYSIS

Region pair	Area Under Curve (AUC)	Asymptotic 95% Confidence Interval	
		Lower Bound	Upper Bound
Frontal-temporal	0.775	0.608	0.941
Frontal-parietal	0.793	0.632	0.954
Frontal-occipital	0.790	0.626	0.954
Temporal-parietal	0.778	0.620	0.935
Temporal-occipital	0.759	0.599	0.920
Parietal-occipital	0.812	0.672	0.951

TABLE II. AREA UNDER CURVE FOR INTRA LOBAR ANALYSIS

Region	Area Under Curve (AUC)	Asymptotic 95% Confidence Interval	
		Lower Bound	Upper Bound
Frontal	0.747	0.570	0.924
Temporal	0.787	0.639	0.935
Parietal	0.769	0.603	0.934
Occipital	0.731	0.567	0.896

Findings of the present work using CRQA method clearly brings out the synchronization differences of the MCI EEG from normal subjects. Also, it highlights the effectiveness of the cross recurrence analysis method in classification of MCI EEG.

IV. CONCLUSION

This work aims at investigation of the effectiveness of CRQA method for MCI EEG classification from EEG of control subjects. This is achieved by analyzing the EEG using CRQA method under memory activation state of picture recollection. In general, higher synchronisation levels are observed in the MCI EEG which is inferred from the higher RR values in both cases of inter and intra lobar analysis. A fair classification between the groups is also obtained as indicated by the AUC values of ROC analysis.

ACKNOWLEDGMENT

The Authors are thankful to Dr. Murali Krishna Menon, Senior Consultant Neurologist of Lakeshore Hospital and Research centre, Cochin, for the support extended during the data collection and helpful discussions for conclusive analysis. Authors would also like to thank the authorities of Welcare hospital, Kochi for their help during data collection. One of the authors, (BMK) would like to acknowledge financial support from Science and Engineering Research Board, Department of Science and Technology (India), through Fast Track Scheme No. SR/FTP/PS-006/2010 and thank Dr. J. Bhatt, Director, STIC, CUSAT for all the support and encouragement.

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